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MSTU 4083 Final Paper - Shanghai Cai Fan (Rice with Salted Pork and Greens) Redesign

I.Stakeholder analysis and problem:

Cooking is a family-based learning domain in which knowledge is often transmitted through everyday participation rather than formal instruction (Rogoff, 2003). The general learning problem is that older adults with extensive home-cooking experience often struggle to engage with online recipes that rely on text-based or app-based formats. This difficulty does not stem from a lack of cooking knowledge, but from a fundamental mismatch between how cooking was originally learned and how it is represented in contemporary digital recipe environments. As a result, conventional online recipes frequently fail to function as accessible or meaningful learning tools for this learner.

1.1 stakeholder analysis:

The primary stakeholder in this study is older adult with extensive home-cooking experience, whose culinary knowledge was acquired through family-based, intergenerational practice rather than formal instruction. For this learner, cooking knowledge is typically developed through participation, observation, and interpersonal interaction within everyday family settings, rather than through self-directed engagement with written procedures.

To ground this analysis in a concrete example, one elderly grandmother was interviewed as a representative case. Over many years, she learned to cook by assisting her husband in the kitchen, where cooking functioned as a relational activity embedded in companionship, care, and shared responsibility. Learning new dishes was therefore experienced as a socially meaningful practice rather than an individual task.

This learning history strongly shapes how such learners encounter modern recipe formats. Many experienced older home cooks therefore avoid written recipe books and applications, not because they lack interest in cooking, but because such formats remove the relational and participatory context that once gave learning its meaning. Reading a recipe is often experienced as completing an isolated task rather than engaging in a shared practice, a pattern commonly observed among older learners whose knowledge is grounded in experience-based and socially embedded learning (Merriam & Bierema, 2014).

In addition, prior learning within this group relies heavily on orality and situated participation. Knowledge is transmitted through verbal explanation, observation, and hands-on

involvement in real kitchen contexts, reflecting forms of tacit and embodied knowledge that are difficult to externalize in written procedures (Polanyi, 1966). When recipes present cooking actions as abstract procedural sequences, new information becomes difficult to relate to familiar routines and lived experience.

Finally, tension arises between externally designed instructional scaffolding and embodied culinary expertise. Having relied on their own methods to nourish families over decades, these learners place greater trust in lived experience than in externally authored recipes. When new methods contradict established routines, learners lack a familiar framework for judging their relevance or legitimacy, resulting in skepticism toward adaptation.

1.2 learning problem:

Based on the stakeholder analysis, the central learning problem lies not in learners' lack of cooking knowledge, but in how current online recipe formats fail to support key learning targets required for this population. Most online recipes implicitly assume individual, text-driven, and procedurally compliant learning, an assumption that conflicts with family-based cooking practices in which learning is shaped through shared activity and lived experience.

At the level of interest, existing recipe formats do not support learners' entry into the learning experience through emotional or relational engagement. For learners whose cooking knowledge was developed through shared practice and companionship, interest emerges through continuity with familiar social meanings rather than through informational completeness alone (Renninger & Hidi, 2015). When recipes are presented primarily as impersonal procedures, learners lack a meaningful affective entry point into learning, which constrains their willingness to invest attention and effort (Deci & Ryan, 2012).

At the level of contextualizing, online recipes represent cooking actions as abstract procedural sequences detached from the perceptual and situational cues learners rely on to make sense of practice. As a result, learners struggle to interpret new steps in relation to familiar routines or to understand why particular variations matter in concrete cooking situations. This difficulty reflects a breakdown in situated sense-making: without access to perceptual cues and practice-based explanations, learners are unable to situate new procedural steps within the experiential routines through which their cooking knowledge was originally constructed (Brown, Collins, & Duguid, 1989).

At the level of attitude, learners are not supported in evaluating new methods in relation to their existing experience. Without opportunities to compare outcomes, interpret differences, or exercise judgment in practice, unfamiliar techniques are encountered as fixed prescriptions rather than adaptable options. This limits learners' ability to develop an open and constructive orientation toward new approaches by constraining their sense of agency and competence during learning (Deci & Ryan, 2012).

II. Our Hypothetical Learning Model and Learning Outcomes

To support the learner's transformation in both engagement and action, we restructure the learning targets and the learning model. Specifically, after participating in this learning experience, the learner can understand cooking as an extension of a close-knit family, thereby developing a sense of value recognition and emotional preference toward cooking. Within a cognitively apprenticed learning mode relevant to her prior experience, the learner can build up a sense of familiarity and belonging, making her more willing to participate in inquiry and hands-on practice **[Engage]**. Meanwhile, upon completing this experience, the learner will be capable of integrating new and existing knowledge through visual comparison, reorganizing her perceptions based on evidence, and transforming insight into effective cooking actions with a receptive manner **[Do]**.

【Interest】First, the learner will be able to embrace new things from the perspective of a family member, evoking the deep emotional memory that "cooking signifies family bonding." By recognizing that learning recipes is not a coercive task but an emotionally meaningful extension of creating a warm and caring household atmosphere, she can rediscover the value of cooking, which ignites a preference for exploring new culinary knowledge.

Constructivism holds that through encountering and comparing different explanations and outcomes, learners can gradually deepen their understanding of a problem or concept (Allen, 2022). Consequently, within a learning environment that encourages multiple perspectives, learners can proactively reflect on and reinterpret the sustained significance of culinary practices in daily life. Ertmer and Newby (2013) further emphasize that once learners gradually discover the significance of inquiry, manipulation, and expression, they will strive to engage in active exploration and experimentation.

【Contextualizing】Next, as the learner begins to actively observe, practice, and test different approaches, She gradually recognizes that this learning environment, which supports

her SRL, is closely aligned with her prior apprenticeship-style home cooking experience. This allows the learner to perceive the relevance of learning new recipes and her previous hands-on learning experiences, gradually fostering a sense of familiarity and belonging. This relevance arises from the reorientation towards a learning ecosystem with which the learner is already familiarized. By constructing a cognitive apprenticeship-based scaffolding environment that integrates modeling, coaching, and feedback, the learner acquires expert-like thinking (e.g., using medium heat first and then simmering on low prevents burning and allows the savory aroma to infuse the rice). Crucially, this learning state, which reactivates familiar experiences and metacognitive knowledge, can promote the learner's autonomy, competence, and relatedness. Consequently, she no longer regards cooking as burdensome and becomes more willing to participate in culinary training.

According to Wang et al. (2017), the social environment is central to the internalization process in OIT. When the environment truly acknowledges learners' choices and emotions, motivation can shift from external regulation to identification and integration, moving learners from forced participation to self-endorsed engagement. Notably, in our culinary instruction that satisfies basic psychological needs, the learner can experience the enjoyment intrinsic to the activity itself. This will soften her rigid adherence to prior knowledge, creating the necessary psychological conditions for subsequent conceptual change.

【Attitude】Ultimately, when encountering new information inconsistent with established knowledge, the learner can identify cognitive conflicts inherent in key steps and reflect on original assumptions based on actual outcomes. For example, after comparing the outcomes of different cooking durations through comparative demonstrations, she recognizes that the assumption "longer cooking yields better flavor" is invalid; instead, the shorter cooking times suggested by the digital recipe more effectively enhance texture and preserve nutrients. Through observation, comparison, and verification, she engages in conceptual change by incrementally revising and restructuring prior beliefs, leading to the development of a flexible schema that integrates new and existing experiences. Ultimately, she can understand the rationale behind each stage and translate understanding into action with a receptive attitude.

In conclusion, by sequentially constructing Interest, Contextualizing, and Attitude, the learner can progress from value recognition and emotional engagement to sustained behavioral change. Based on this, this paper proposes the following practical design solutions.

III. Design:

Based on the three targets identified through the analysis of the learner's problems, the overall design of the Shanghai Cai Fan recipe will introduce an AI virtual avatar system. This AI avatar will simulate the learner's family member and provide one-on-one customized guidance. Once generated, the AI avatar will participate throughout the entire recipe page. It will automatically play and display content without requiring any additional clicks from the learner. This design logic aligns with LXD, which stresses designing learning experiences from the learner's perspective. It requires understanding the learner's background, context and helping them construct meaning through personal and social experiences (Chang & Kuwata, 2020). Therefore, the appearance of a relative-like AI Avatar can create a familiar and emotionally meaningful recipe-learning process, which helps learners stay engaged.

Detail Design 1: AI Avatar for Recipe Storytelling

This detailed design aims to address the lack of **interest** among the learner in learning new recipes. And it will be placed at the beginning of the recipe. To achieve this goal, the design relies on speech recognition technology, AI-driven character generation technology, and emotional storytelling algorithms, which collaboratively support personalized interactions between learners and the recipe.

To re-spark the learner's interest, this detailed design will automatically activate the voice-enabled AI Avatar when she accesses the recipe. The AI Avatar will first ask which family member the learner wishes it to simulate. Based on her description, the AI Avatar will generate a matching avatar. Then, the personalized AI Avatar will narrate warm cooking-related stories that connect with the learner's emotional memories. This helps re-engage the learner and keep her to continue staying on the page and following the recipe.

For example, in this design, when the learner clicks into a recipe, the AI Avatar will ask her a personalized question. It might say, "Whose image would feel most familiar to you when learning the recipe tutorial?" Based on her description, the AI Avatar will then transform into a virtual version of the relative, such as the AI Avatar grandpa. From a relative's perspective, the AI Avatar grandpa will explain to the learner why she should embrace the new online format of the Shanghai Cai Fan recipe. He will gently share emotionally resonant memories, such as: "Dear, learning new recipes can still bring the whole family together to cook, just like when we cooked together in the past." By watching the AI virtual grandpa share this heartwarming story

about cooking, the learner will rediscover that learning new recipes can still bring the family together. The AI Avatar can ultimately evoke the feeling of cooking with loved ones from her memories. This connection helps the learner rebuild a sense of relatedness, thereby creating the motivation to continue learning this online recipe.

This detailed design is based on the concept of relatedness in Self-Determination Theory (SDT). As Deci and Ryan note, supporting the learner's relatedness is essential for internalizing new behaviors (2012). They describe relatedness as "the need to be close to, trusting of, caring for, and cared for by others," which connects to the human "need to belong" (Deci & Ryan, 2012). In this design, the AI Avatar appears in a familiar way, helping the learner feel emotional support and closeness when she enters the recipe. This strengthens her sense of relatedness. Since "motivation is the most important factor that drives learning" (Gee, 2003), it also becomes essential to create conditions that strengthen the learner's motivation to stay engaged. Enhancing her sense of relatedness and interest can support this process. As a result, these factors can help the learner engage with the online Shanghai Cai Fan recipe study.

Detail Design 2: AI Avatar "Word-of-Mouth" Tutorial

This detailed design aims to address the learner's difficulty in engaging with new online recipes due to the lack of contextualized learning in online recipe formats. While the learner has strong prior cooking experience, she struggles to connect new recipes with her existing mental models and lived experience. For the learner, cooking knowledge has always been acquired through socially situated, oral, and interactive practices rather than through decontextualized written instructions. To solve this problem, this design relies on continuous voice interaction, eye-tracking support, and adaptive dialogue generation to recreate a familiar "word-of-mouth" learning environment through an AI Avatar embedded throughout the recipe process.

To restore this contextualized learning experience, the AI Avatar remains active throughout the entire recipe tutorial and engages the learner in ongoing, back-and-forth dialogue. Instead of presenting fixed procedural steps, the AI Avatar translates each cooking action into conversational, everyday language that mirrors the neighborhood-style teaching the learner is familiar with. The learner can speak to the AI Avatar at any time to ask questions, confirm steps, or express uncertainty. The AI Avatar responds in a supportive, oral manner—sometimes even using dialect-based phrasing—so the learner feels she is continuing her original learning culture rather than being forced to adapt to a foreign digital format.

For example, when the learner begins cooking the Shanghai Cai Fan recipe, the AI Avatar (such as AI Avatar grandpa) may say, “Now we’ll heat the pan first—just like how I used to check the oil temperature before adding the rice.” As the learner proceeds, the system detects where her eyes are focused and automatically triggers the next spoken instruction, eliminating the need to scroll or tap. If the learner makes a mistake—such as adding water before the oil is fully heated—the AI Avatar does not interrupt or signal failure. Instead, it gently reassures her by saying, “It’s okay, this happens often. Let’s just lower the heat a bit now—the rice will still absorb the flavor.” Through this dialogue, the learner realizes that mistakes do not ruin the entire process and that guidance is available without disrupting her independent learning flow. This continuous conversational support allows the learner to remain actively engaged as a participant in the recipe rather than a passive follower of instructions.

This detailed design is grounded in multiple learning theories. First, from the perspective of SDT, the AI Avatar supports both relatedness and competence by maintaining social presence while helping the learner feel capable of managing each step of the recipe (Deci & Ryan, 2012). Second, the design aligns with principles of Self-Regulated Learning (SRL), as the learner monitors her progress, requests assistance when needed, and adjusts her actions without external pressure or rigid pacing (Winne & Azevedo, 2022). Finally, from a constructivist perspective, learning is enabled through social negotiation, where understanding emerges through dialogue, explanation, and shared meaning-making rather than through isolated information delivery (Brown, Collins, & Duguid, 1989). Together, these mechanisms allow the learner to meaningfully reconnect new recipe learning with her prior cooking experiences.

Detail Design 3: AI Avatar Step-by-Step Gif/Images

Since learning **attitudes** influence whether the learner is willing to explore new recipes, the design focuses on making recipes sound more approachable and credible. The third design goal is to encourage the learner to try new online recipes rather than reject them due to past experiences, health concerns, or distrust of novel recipes. To address these issues, the goal decided to create a new design based on the AI Avatar. When the learner poses questions, the AI won’t just respond with text. Instead, it will automatically generate two comparative GIFs or images: one showing the method for the new online recipe, and another depicting the familiar traditional one (Tversky, 2011).

For example, when the learner asks: “Is this recipe healthy? Why does this recipe only take 13 minutes, but based on my experience, I need 20 minutes to make it?” The AI Avatar immediately displays two animated gifs: one showing the slow version and another showing a quicker, lighter, less oily, healthier version, explaining why the times differ. Through this comparison, the learner clearly sees the differences between the two approaches and understands that the new recipe isn’t dismissing her past experience but rather adjusting it based on her health needs (Mayer, 2017).

The entire process makes the learner feel that this recipe is tailored just for her, rather than forcing her to accept other methods. This approach not only alleviates her reservations about the new recipe but also encourages her to proactively learn and experiment with new cooking techniques (Van Merrienboer & Sweller, 2005). In this way, the AI Avatar acts like a patient family member offering guidance beside the learner. When the learner thinks: “I already know how to make this. Why should I follow someone else’s method?” The AI Avatar then will use GIFs or images to compare two version recipes. The learner doesn’t have to guess the differences or feel pressured to abandon her old ways. Instead, the learner instantly sees: the new version just makes small changes to help her eat healthier.

With this approach to explanation, the learner will feel that the content genuinely relates to her, rather than feeling like she is reading someone else’s recipe book. She becomes more open to recognizing that even if she already knows how to make a dish, a new version may better fit her current physical condition or daily needs. This design is also supported by the SDT concept of autonomy, because the learner is free to compare the two versions, make her own choices, and form her own understanding without feeling pressured or controlled. At the same time, by clearly showing the differences between old and new methods, the design encourages conceptual change: it helps the learner revise her earlier belief that “my way is the only correct way,” and gradually accept that updated techniques can coexist with, or even improve upon, her traditional knowledge. Together, these elements naturally encourage the learner to continue using the online recipes and to try out healthier and new methods at her own pace.

IV. Assessment:

To evaluate if the *AI Avatar for Recipe Storytelling* successfully stimulates the learner’s interest in the new online version of the recipe, the assessment will examine whether the learner prefers to continue learning the recipe over quitting after interacting with the AI Avatar. As

Renninger and Hidi stated, learners' effective interest activation depends on the alignment between their characteristics and external stimulation (2015). Therefore, this evaluation will examine if the emotional narratives and personalized AI Avatar truly meet the learner's needs and thereby increase the learner's interest in this recipe learning.

Firstly, after the AI Avatar completes the emotional storytelling, this assessment will immediately pose a series of brief questions to the learner through voice prompts. For example, the AI Avatar will ask the learner, "Did this story make you more eager to continue learning the recipe?" These questions aim to detect whether her interest has been triggered. Then, the AI Avatar will track the learner's actions after hearing the narrative. It begins by observing the learner's actions, such as whether she stays on the recipe page or moves to view ingredient preparation. These behaviors can reveal whether the learner's interest has been aroused or not. They also show if this interest has eventually led the learner to continue studying the recipe.

In this assessment, when the learner provides positive responses in self-reports, it shows that the story enhanced her willingness to continue learning the recipe. This indicates that the design is successful. The success is further supported by the learner's behaviors. For example, after hearing the narratives, the learner chooses to remain on the recipe page or even begins to review ingredient preparation. These behaviors show that the learner's interest has been sparked and beginning to sustain. All those reactions can indicate that the design of the AI Avatar for Recipe Storytelling is successful. Oppositely, if the learner self-reports that the story was unhelpful, this indicates that she still lacks interest. If the learner even immediately exits the page or refuses to proceed to the next step, it further supports the failure of this design. In a word, these reactions indicate that the emotional narrative failed to connect with the learner's experiences.

Based on the evaluation results, this design also incorporates corresponding remedial measures. First, the AI Avatar system will more actively engage with the learner by using gentler emotional conversations. For example, the AI Avatar might ask, "Would you like to share a memory related to cooking with me?" By encouraging the learner to express personal stories, the AI Avatar can then capture more personalized emotional cues. After that, the AI Avatar will regenerate a new avatar image that aligns more closely with the learner's emotional memories based on the shared recollections. This can further enhance the learner's emotional identification and trust. By doing so, it can also strengthen the learner's sense of relatedness and interest.

Moreover, the system will also provide alternative emotional narrative versions. For instance, if the initial narrative fails to ignite interest, the system may switch to shorter, more companionable storytelling styles or incorporate more familiar domestic imagery. This helps the learner reconnect the act of "learning to cook" with warm family experiences.

The assessment for the design of *AI Avatar "Word-of-Mouth" Tutorial* focuses on whether the learner is able to meaningfully connect learning with her existing mental models and lived experience. Rather than evaluating procedural accuracy alone, this assessment attends to how the learner interprets, adapts, and reflects on the recipe as part of a socially situated practice. Meaningful learning, in this context, is evidenced by the learner's ability to explain cooking decisions in her own words, relate new steps to prior family-based practices, and express confidence in navigating variations of the recipe.

To assess this, the learner will be prompted at natural pause points to verbally reflect on her actions. For example, the AI Avatar may ask: "Why do you think we switched to a lower heat here?" or "How does this step compare to how you used to cook with your grandpa?" These responses allow evaluators to observe whether the learner is integrating new knowledge into her existing framework rather than merely following instructions.

Success is indicated when the learner demonstrates the ability to articulate reasoning, draw parallels to past experiences, and proceed through the recipe with reduced hesitation and increased autonomy. Success is indicated when the learner demonstrates the ability to articulate reasoning, draw meaningful parallels to past experiences, and proceed through the recipe with reduced hesitation and increased autonomy.

To determine whether *AI Avatar Step-by-Step Gif/Images* design genuinely helps the learner adopt new online recipes, evaluate based on several observable behaviors. The key is to see whether the learner's attitudes, along with the manners in which she responds, begin to reflect alignment with the new recipe and show that it fits her existing thoughts, opinions, and cooking beliefs.

First, observe whether the learner begins to ask questions of the AI Avatar and view comparison gifs or images. If she engages with this content, it indicates she's moving beyond relying solely on past experience and starting to wonder how the two methods differ, this is a crucial first step toward accepting new approaches.

Next, observe whether the learner is willing to attempt cooking a dish following the steps in the updated recipe. She doesn't need to replicate it perfectly on the first try, simply being willing to follow the new steps is a clear sign of progress. It indicates she is no longer resistant to recipes written by others and is open to seeing if these adjustments are genuinely healthier or better suited for her.

If these behaviors occur, it signifies the design has successfully motivated the learner to try the new recipe. It also confirms that the entire interactive approach has effectively strengthened her willingness to learn new methods.

If the learner still struggles to adopt new recipes when using this design, likely due to excessive cognitive load (Tversky, 2011). When information volume exceeds her capacity, it triggers rejection, confusion, or abandonment. This means the design has failed. Even though the new approach minimizes text, the learner perceives the two versions of the animated graphics, time differences, and health explanations as complex content. Excessive comparisons may heighten her stress, discouraging continued use. The goal then should shift to gradual acceptance: the system could reduce information density by focusing on one key point at a time, allowing her to experiment at a more relaxed pace. For example, focus on only one step at a time. When asked about the difference between cooking the same vegetable for 13 minutes versus 20 minutes, the AI Avatar will generate only two comparison images: the greener one shows the vegetable cooked for 13 minutes, while the yellow one shows it cooked for 20 minutes. Beyond that, avoid excessive information and generate just two images highlighting the key points of a single question.

V. Reflection:

The Challenge of Accurately Identifying Learning Problem and Learning Target

In our initial outline, the Learning Problem associated with the contextualized target did not accurately capture the learner's core learning barrier. At the time, our description placed excessive emphasis on the fact that the recipe failed to provide a contextualized and interactive learning environment. However, this framing pointed more toward a potential solution than toward the learner's actual difficulty.

The true underlying learning problem is that the text-based recipe presents the cooking process in a procedural and decontextualized manner, which does not naturally align with the learner's prior cooking learning mode rooted in cognitive apprenticeship.

Upon reflection, we realized that our earlier articulation of the Learning Problem lacked precision and focus, preventing us from identifying the learner's genuine constraint. Moving forward, we plan to avoid such ambiguous phrasing and instead focus on clearly and accurately defining Learning Problems and Learning Targets, thereby providing a stronger foundation for subsequent design decisions.

The Challenge of Establishing the Design Framework

In the initial design stage, we encountered a significant disagreement in our design. At the beginning of the process, we originally intended to develop three completely distinct directions based on the three identified targets. However, we later discovered that although we still generated three independent targets, each target was closely interconnected. This interdependence made it impossible for our designs to function as three entirely separate and distinct concepts. We then had a long meeting to discuss the solution. In that meeting, we finally summarized our design ideas and realized that all our concepts were refined designs based on the AI Avatar system. This led us to develop an overall design framework, which was to introduce the AI Avatar system. After that, we further refined three smaller designs based on the specific problems identified by the stakeholder (the learner/grandma) and the corresponding targets derived from her.

Therefore, in the design section, the key difference between this paper and our midterm paper became clear. In this paper, we introduced the AI Avatar system design model at the beginning of the design part and provided a brief explanation of it. We believed this approach helped readers better understand our subsequent designs. Additionally, since our three detailed designs were more refined versions based on the recipe itself and the AI Avatar, we felt it was essential to include an introduction to the overall design before explaining the detailed designs. As a result, our final design section included an initial overview of the overall design followed by the three detailed designs.

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